BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

LESSON 253 · Clock speed, cache and pipelining

Name: _____ Class: _____ Date: _____

Work through the pages in order. Answer in BLACK PEN first. Check in GREEN PEN only after the Exit Ticket.

Do Now · Recall previous learning

Q1. What does the data bus transport during an instruction fetch?

Q2. Which register holds the address of the next instruction?

Q3. What is a dependency between tasks?

Q4. A rising-edge D flip-flop initially has Q=0. D changes 0→1 between edges, then 1→0 just before the next rising edge. What is Q immediately after that edge? Justify.

BLACK PEN Answer all questions first. GREEN PEN Then: tick and rewrite mistakes.

I Do · Read new knowledge

Key vocabulary

clock speed — Number of processor clock cycles each second.

cache — Small, fast memory that reduces some main-memory accesses.

pipeline — Overlap stages of several instructions.

hazard — Dependency or uncertainty disrupting smooth pipelining.

throughput — Amount of work completed in a given time.

New learning

Different performance factors — Clock speed indicates cycles per second; multiple cores can execute independent work; cache reduces some slower-memory access.

Overlap pipeline stages — Without overlap, three three-stage instructions need nine stage slots; an ideal pipeline finishes them by the fifth cycle.

Account for hazards — Branches, data dependencies and waiting for memory may cause stalls or pipeline flushing; pipelining does not make every instruction individually three times faster.

Worked example

Trace two independent instructions through a 3-stage F–D–E pipeline.

1. Predict first — Cycle 1 fetch I1. In cycle 2 decode I1 while fetching I2.

2. Explain and check — Cycle 3 executes I1 and decodes I2; I2 executes in cycle 4.

We Do · Practise with guidance

Trace four independent instructions through the same pipeline. In which cycle does I4 finish?

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3 stages; 4 independent instructions.
Ideal completion cycle = 3+4−1=6
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Predict first

Begin a new fetch on each clock cycle, while older instructions advance to Decode or Execute.

Explain and check

Instruction four starts Fetch in cycle 4, Decodes in 5 and Executes in 6.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

You Do · Work independently

BLACK PEN: Write your own answers. Do not look at the Review & Improve pages yet.

Question 1

State two processor performance factors and explain one way each might improve throughput.

Question 2

For a three-stage F–D–E pipeline with three independent instructions, write the stages occupied in cycles 1–5 and the completion cycle for I3.

Question 3

Explain why doubling clock speed does not necessarily double an application's performance. Include one pipeline hazard.

Exit Ticket · Answer before checking

Pipelining always reduces the number of stages required for an individual instruction.

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Review & Improve · GREEN PEN

Now change to GREEN PEN. Find each original question. Tick correct answers; read the model and rewrite mistakes.

Page 1 onwards · Do Now

Q1. The instruction's contents from memory towards the CPU.

Q2. PC.

Q3. One operation cannot proceed until another result is ready.

Q4. Q=0. The flip-flop samples D at the rising edge (D=0 at that instant); an earlier pulse between clock edges is not stored.

You Do pages · Independent questions

Q1. [4] Higher clock rate may allow more cycles each second [2]; larger/faster cache can reduce average memory latency [2]; alternatively cores can run independent tasks.

Q2. [6] C1 I1:F [1]; C2 I1:D, I2:F [1]; C3 I1:E, I2:D, I3:F [2]; C4 I2:E, I3:D [1]; C5 I3:E [1].

Q3. [6] Memory stalls/cache misses [1], non-parallel parts of software [1], instructions per cycle vary [1], branch hazards can flush a pipeline [1], data dependencies stall instructions [1]; measured speedup depends on the bottleneck [1].

Exit Ticket · Model answer

False. It usually overlaps stages to improve throughput, but each instruction still passes through its stages.

Try the question again

Choose one question you got wrong. All correct? Choose the hardest.

1. Copy the whole original question:

2. Write a complete new answer in GREEN PEN. Show working if needed:

BLACK PEN Answer all questions first. **GREEN PEN** Then: tick and rewrite mistakes.

Do Now Recall previous learning | **I Do** Read new knowledge | **We Do** Practise with guidance | **You Do** Work independently | **Review & Improve** Check understanding